sg13g2_stdcell_hv — Generation of a Thick-Oxide (3.3 V) Standard Cell Library

IHP SG13G2 open PDK · derivation, all views, verification and physical sign-off

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1 Scope and result

The IHP SG13G2 open PDK ships one digital standard cell library, `sg13g2_stdcell`, built entirely from the 1.2 V thin-oxide devices (`sg13_lv_nmos` / `sg13_lv_pmos`). Its 3.3 V thick-oxide devices appear only inside the `sg13g2_io` pad ring — 22 macros, all pads, fillers and a corner — so nothing at 3.3 V can be synthesised and placed.

`sg13g2_stdcell_hv` fills that gap: **all 84 cells** of `sg13g2_stdcell`, rebuilt on `sg13_hv_nmos` / `sg13_hv_pmos`, with the same topology, the same pin names and the same pin order. Cells are named `sg13g2_hv_*` so the 1.2 V and 3.3 V libraries coexist in one netlist. This report documents how every view was generated, how each was verified, and the final sign-off state.

Deliverable	Coverage	Status
SPICE netlist (`spice/`)	84 cells, 924 devices	verified against 3 independent views
CDL netlist (`cdl/`)	84 cells	LVS reference, all 66 drawn cells match
Verilog (`verilog/`)	84 modules	shared `ihp_*` UDPs, deliberately not duplicated
xschem symbols (`sym/xschem`)	84	copies of the thin-oxide symbols
xschem schematics (`sch/xschem`)	84 + gallery sheet	netlist-equivalence proven
GDS layout (`gds/`)	66 of 84 cells	**DRC clean, LVS clean**
LEF abstracts (`lef/`)	66 macros + `CoreSiteHV` site	generated from the GDS, pin sets verified against CDL
Liberty NLDM (`lib/`)	66 cells, 600 timing tables	combinational + sequential (clk→Q, setup/hold); areas from layout

Deliverable	Coverage	Status
Registration (`xschem_lib_*.tcl`)	library path + gallery hook	idempotent

Physical sign-off, from a single fixed invocation of the PDK's own klayout decks on an abutted two-row array of all 66 drawn cells:

Check	This library	IHP thin-oxide control (same harness)
DRC, cell rules (`Act.*`, `Gat.*`, `M1.*` spacing/width, `TGO.*`, `NW.*`, `Cnt.*`, …)	**0**	0
DRC, chip-level metal density (`M1.j` … `TM2.c`)	7	6
LVS	**66** / **66 match**	fill cells fail identically without the documented relaxation

The density items are properties of the test harness, not of either library: a standard cell library contains no Metal2 and above, and metal density is a check on a *filled* die. The one count that differs, `M1.j` (min 35 % Metal1), is quantified rather than waved away: the deck measures over the marker bounding box including the empty ThickGateOx margin, where the padded 17-track array reads 35.0 % — over the actual cell area it is 37.1 % — while the denser, unpadded thin-oxide control reads 45.2 % and so never trips it. A placed block carries metal fill and does not see this.

2 The device transform

Applied to every logic device in all 84 cells:

	thin oxide	thick oxide
device model	`sg13_lv_nmos` / `sg13_lv_pmos`	`sg13_hv_nmos` / `sg13_hv_pmos`
gate length	130 nm	**450 nm** (`Gat.a3` minimum for the 3.3 V FET)
NMOS width	—	unchanged
PMOS width	—	**× 2.40**, snapped per finger to the 5 nm grid
`as` / `ad` / `ps` / `pd`	—	recomputed from device geometry
supply	1.2 V	3.3 V

Gate lengths already at or above 450 nm were left alone, because they were chosen deliberately rather than set by the minimum: the `decap_4` / `decap_8` MOS capacitors (1.0 µm), the `sighold` keeper devices (700 nm) and `dlygate4sd3_1` (500 nm). 914 devices had their length raised. The delay cells `dlygate4sd2_1` (180/250 nm) and `o21ai_1` (150 nm) sat below the thick-oxide minimum and had to move to 450 nm, which shifts their delay ratios relative to the thin-oxide originals. `sg13g2_hv_antennanp` is unchanged apart from its name — it contains `dantenna` / `dpantenna` junction diodes, to which gate oxide thickness does not apply.

2.1 Why PMOS × 2.40

The thin-oxide `inv_1` uses $W_p/W_n = 1.12/0.74 = 1.5135$. Simulation shows this is *not* a drive-matched ratio: at 1.2 V the pair gives $I_{dsat,p}/I_{dsat,n} = 0.82$, and equal drive would need $W_p/W_n = 1.894$. What 1.5135 does give is a switching threshold at

$$V_m/V_{DD} = 0.5046$$

essentially mid-rail. The thin-oxide library is sized for a centred switching threshold, and that is the property carried over.

At 3.3 V with $L = 450\,\mathrm{nm}$ the thick-oxide PMOS is much weaker relative to its NMOS (219 µA/µm against 533 µA/µm), and reaching the same $V_m/V_{DD} = 0.5046$ needs $W_p/W_n = 3.63$ — a factor

$$K_p = \frac{3.63}{1.5135} = 2.40$$

on the thin-oxide ratio. Applying K_p to every PMOS and leaving every NMOS alone preserves each cell's internal stack ratios — where the thin-oxide library encodes its series-device compensation — while re-centring the switching threshold. Checked across the library's full 0.3–17.9 µm width range, the required ratio varies between 3.63 and 3.97, so one global factor holds to about ±5 %.

Method: a self-biased inverter (input tied to output) settles at exactly V_m, so one operating point per candidate ratio gives the answer with no sweep interpolation (`work/vm_sweep.py`, `work/ratio_check.py`).

2.2 Parasitic recomputation

Source/drain areas and perimeters follow the vendor deck's own geometry formulas. With finger width $w_f = \max(W/n_g, 0.15\,\mu\mathrm{m})$, $z_1 = 0.34\,\mu\mathrm{m}$ and $z_2 = 0.38\,\mu\mathrm{m}$, an odd finger count gives

$$A_S = A_D = w_f\left(z_1 + \frac{n_g - 1}{2} z_2\right)$$

and for even n_g the source and drain differ:

$$A_S = w_f\left(2z_1 + \frac{n_g - 2}{2} z_2\right)$$

$$A_D = \frac{w_f z_2 n_g}{2}$$

with the matching perimeter expressions. Before any generation the formulas were validated by recomputing all 924 thin-oxide devices and comparing against the shipped values: worst relative error 3.75×10^{-4}, entirely explained by the vendor netlist printing four significant figures. The generator refuses to run if this validation fails.

2.3 What the transform costs

Measured on `inv_1` versus `sg13g2_hv_inv_1` (`work/fo4.py`):

	thin oxide @ 1.2 V	thick oxide @ 3.3 V	ratio
input capacitance	2.67 fF	5.87 fF	2.20×
FO4 delay	53.6 ps	142.4 ps	2.66×

The thick-oxide NMOS delivers *more* current per micron at 3.3 V than the thin-oxide one at 1.2 V (533 vs 391 µA/µm) — the higher supply more than compensates for the 3.5× longer channel. The speed loss is capacitance, not drive.

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3 Netlist generation

`work/gen_hv_lib.py` parses the PDK's thin-oxide netlists and re-emits every view from one internal model, so a device cannot silently diverge between views. Regex matching uses named groups throughout — an early version with positional groups shifted trailing indices and duplicated schematic attributes, a class of bug the named form makes impossible.

Order of operations per device: substitute the model name, raise L, scale the PMOS W, recompute `as/ad/ps/pd` from the new geometry, round consistently. The generator validates the parasitic formulas against all 924 source devices before writing a single output file.

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4 Deliverables, view by view

4.1 `spice/sg13g2_stdcell_hv.spice` — simulation netlist

1 536 lines, 84 `.subckt` blocks, 924 devices as `X`-cards calling the PSP103 Verilog-A models (loaded via OSDI in ngspice). This is the view every functional verification in section 5 runs on. Device widths and lengths are synchronised to the drawn layout for the 66 cells that have one (section 6.6); the antenna cell's diode `w/l/a/p` follow the drawn diode.

4.2 `cdl/sg13g2_stdcell_hv.cdl` — LVS netlist

1 783 lines, the same 84 subcircuits as `M`-cards with `*.PININFO` directives, consumed by the PDK's klayout LVS deck. Kept separate from the SPICE view because the two serve different tools, but generated from the same internal model and compared field-by-field by `verify_sch.py`.

4.3 `verilog/sg13g2_stdcell_hv.v` — functional view

3 273 lines, 84 modules renamed to `sg13g2_hv_*`. The file deliberately does **not** ship a copy of `sg13g2_udp.v`: the user-defined primitives are named `ihp_*` and shared with the thin-oxide library, and a second copy in one elaboration would collide.

4.4 `sym/xschem/*.sym` — 84 symbols

Copies of the thin-oxide symbols: same drawn geometry, same pin names, same pin order, so schematics port between the libraries by netlist prefix alone. Two things differ: the template prefix is `sg13g2_hv_`, so an instance netlists to `sg13g2_hv_<cell>`, and the symbols carry `type=subcircuit`, resolving their schematic through `$::SG13G2_HV_SCH` — the thin-oxide symbols instead resolve through the PDK's `hierarchy_config` proc, which is hard-wired to the thin-oxide schematic directory and has no thick-oxide view.

4.5 `sch/xschem/*.sch` — 84 schematics and the gallery

One schematic per cell, netlisted through the symbols by `verify_sch.py` so that symbol→schematic resolution is exercised, and compared against SPICE and CDL on pin lists and every device's model, width, length, finger count and multiplier: **84/84 cells, 924 devices, three views agree**. The single intentional divergence — `dlygate4sd2_1`'s PMOS drawn at $L = 0.625\,\mu$m (section 6.1) — is carried consistently in all three views.

`sg13g2_hv_stdcells.sch`, in the same directory, is the gallery: all 84 cells on one sheet in 13 functional groups, the thick-oxide counterpart of the PDK's `IHP130_stdcells.sch`. It is generated by `work/gen_gallery.py`, which computes each symbol's drawn extent from its primitive records and packs the groups without overlap whatever the pin count. It lives in `sch/xschem/` so it resolves by name like any cell, and project `xschemrc` files open it at startup via `$::SG13G2_HV_GALLERY`.

4.6 `xschem_lib_sg13g2_stdcell_hv.tcl` — registration

Sourced after the PDK `xschemrc`, it appends `sym/xschem` and `sch/xschem` to `XSCHEM_LIBRARY_PATH` (deduplicated — double-sourcing is safe), sets `$::SG13G2_HV_SCH` for symbol resolution and exports `$::SG13G2_HV_GALLERY` for startup hooks.

4.7 `gds/sg13g2_stdcell_hv.gds` — layout, 66 of 84 cells

Produced by `work/layout_retarget.py` from the thin-oxide GDS; the generation method and its sign-off are the subject of sections 6 and 7. Row height is a uniform 7.140 µm — 17 horizontal routing tracks, up from the thin-oxide 3.780 µm / 9 tracks — every cell width is a multiple of the 0.48 µm site, the 66-cell test row is 430.56 µm wide, and all 5 082 contacts are exactly 0.16 × 0.16 µm. The 18 cells without layout are those whose thin-oxide geometry leaves no room for the thick-oxide well clearances at a shared row height (section 9).

4.8 `lef/sg13g2_stdcell_hv.lef` — placement abstracts

66 macros plus the `CoreSiteHV` site (0.48 × 7.140 µm), generated from the drawn GDS by `work/gen_lef.py`. Pin ports are the connected groups of the Metal1/Metal2 pin datatypes, labelled by the text each group contains (`sdfbbp_1`'s D pin routes through Via1 to Metal2 and is emitted on all three layers); the OBS blocks are the drawn metal minus the pin geometry, decomposed at region level so a pin cut out of a strap leaves a hole rather than being covered. Pin `DIRECTION`/`USE`/`NETEXPR` and macro `CLASS` (SPACER for fill/decap, ANTENNACELL for the diode) are inherited from the thin-oxide LEF, whose pin names this library shares by construction — but antenna values are **recomputed from the netlist**, since the thin-oxide numbers are wrong for 3.5× longer gates: `ANTENNAGATEAREA` is the summed W × L on the pin's gate net, `ANTENNADIFFAREA` the summed `ad`/`as` on its drain/source nets. Generation asserts every height equals the site, every width is a site multiple and every pin set matches the thin-oxide reference; the emitted file is then parsed back with klayout's LEF reader and its pin sets verified against the CDL (66/66).

4.9 `lib/sg13g2_stdcell_hv_typ_3p30V_25C.lib` — Liberty NLDM

52 combinational cells at the 3.3 V / 25 °C typical corner, characterised with CharLib against the final, layout-synchronised SPICE netlist (section 8): 25 839 combinational measurements plus a 2 962-task sequential run, 600 timing tables across 66 cells, none empty.

4.10 `doc/`, `README.md`, `work/`

`doc/sg13g2_stdcell_hv.celllist` is the machine-readable cell inventory, and this report lives in `doc/report/`. The `README.md` carries the full engineering narrative including every measurement caveat. `work/` holds the 19 generator, verification and sign-off scripts plus the ngspice OSDI shim — the complete provenance: nothing in the library was drawn or edited outside these scripts except the five documented Metal1 re-routes, which are themselves data in `layout_retarget.py`.

5 Functional verification

The functional suites use the thin-oxide library as the golden reference: the transform changed devices and voltages, never logic, so any topology error appears as a divergence. No truth table was written by hand.

Suite	Coverage	Result
`verify_logic.py`		**PASS**

Suite	Coverage	Result
	60 combinational cells, 452 vectors, both libraries in one deck, outputs digitised	
`verify_seq.py`	16 stateful cells, 400 clocked samples from a common reset	**PASS**
`verify_sch.py`	84 schematics + CDL vs SPICE, 924 devices field-by-field	**PASS**
LVS (klayout)	66 drawn cells vs CDL	**66/66**

Details that matter: the 12 high-impedance states of the tri-state cells (`ebufn_*`, `einvn_*`) are exercised, with a 1 GΩ resistor to mid-rail keeping a floating output distinguishable from a driven 0. Stateful cells are identified from the thin-oxide Liberty by `ff()`, `latch()`, `statetable()` *or* `clock_gating_integrated_cell` — the last two matter because `lgcp_1` / `slgcp_1` are bistable integrated clock gates without an `ff`/`latch` group, and a DC sweep on them is meaningless. Samples with `SET_B` and `RESET_B` asserted together are illegal states with no defined reference value and are excluded rather than counted as passes. Both suites were re-run after the last netlist synchronisation and pass on the shipped files.

6 Layout generation

The thin-oxide GDS is hand-drawn 2-D layout, not a regular template, so a placement-based generator would have to rebuild every cell — and would not be LVS-correct without a router. What can be done *exactly* is a **1-D retarget**: monotone piecewise-linear coordinate maps applied to every vertex. Monotone maps preserve topology, so connectivity survives by construction; the engineering is in choosing the breakpoints and in the exceptions.

6.1 The maps

In **x**, gates must widen from 0.13 to 0.45 µm about their centres with a fixed slope L_{hv}/L_{lv} — a variable slope staggered multi-finger gates to 0.28/0.39 µm. Long-channel gates (the deliberate 0.5–1.0 µm devices) are only widened if below 0.45 µm. The contacted poly pitch grows from 0.48 to 0.80 µm, keeping the thin-oxide gate-to-contact budget around the longer gate. One consequence is accepted rather than fought: `dlygate4sd2_1` staggers an 0.18 µm NMOS gate against an 0.25 µm PMOS gate at overlapping `x`, and a single monotone map cannot widen two overlapping intervals by different factors — its PMOS lands at a legal 0.625 µm, carried consistently into all netlist views.

In **y**, the PMOS band is scaled ×2.40 and three channel inserts open the thick-oxide clearances: 0.215 µm for `NW.d1.dig` (NWell to N+ Activ, 0.31 µm inside ThickGateOx with DigiBnd), 0.100 µm for `pSD.j1`, and 0.430 µm at the rail so the VSS-rail p+ tap keeps 0.40 µm (`pSD.i1`) to the NMOS gates. The band is derived **library-wide** (1.925–3.630 µm, cut at y =

1.595), not per cell — a per-cell band produced six different row heights, which is not a standard cell library. Everything on every layer moves through the maps, including Metal2, Via1 and marker layers: leaving any layer behind silently corrupts the cell (it clipped the antenna diode's extracted area before that was made unconditional).

6.2 PMOS diffusion: slab decomposition

Scaled widths must be exact, and $2.40 \times 5\,\mathrm{nm} = 12$ nm does not land on the 5 nm grid, so snapping the two Activ edges of a device independently gives the same thin-oxide 1.12 µm device 2.685 µm at one y and 2.690 µm at another — and the netlist cannot say which is which. `retarget_pmos_activ` therefore cuts each PMOS Activ polygon into vertical slabs at every vertex x and re-emits each slab at $[\,y_{top} - \mathrm{snap}(2.40\,h),\ y_{top}\,]$. Three rules there were each learned by breaking them:

- **Per connected piece, not per slab bounding box.** A non-convex Activ can hold two disjoint lobes at the same x; the bounding box spans the gap and merges two channels — which silently deleted two PMOS from `dfrbp_1`.
- **Per slab, not per polygon.** A notch is how this library gives devices in one diffusion different widths; forcing a polygon to one height collapses them (`and2_1`'s three PMOS all became 2.69 µm).
- **Sub-grid step alignment, with channel slabs frozen.** The source's own 5 nm jogs land neighbouring slabs one grid step apart, and each such staircase edge within 0.23 µm of a gate edge is an `Act.c` violation (a 0.12 µm slab flanked by steps was an `Act.a`). Aligning *everything* removed them but moved channel edges: two formerly identical devices became 2.400 and 2.405 µm, and since the netlist synchroniser pairs devices to channels by sorted width — ambiguous between near-equal devices on different nets — five cells lost their LVS net correspondence. The final pass freezes every slab that carries gate poly and moves only source/drain slabs toward their channel neighbours, so it provably cannot change any device.

6.3 Contacts and vias

Cut layers are fixed-size and are **translated, never scaled** — a stretched via fails its own width rule. Contacts are removed before the map and re-tiled inside the new Activ at a 0.36 µm pitch (not the thin-oxide 0.34 µm: `Cnt.b1` raises required spacing to 0.20 µm once an array exceeds 4 × 4, which the 2.4× taller PMOS contacts now do), grouped by the polygon they sit in rather than by x (the VSS and VDD rail taps share an x, and x-grouping tiled contacts straight up through open field), each box built by snapping one corner and adding the size (snapping both edges independently rounds 0.16 µm to 0.155/0.165). An epsilon guards the keep test — `hi - lo = 0.15999999999999998` once dropped every rail-tap contact and with them the substrate tie.

6.4 ThickGateOx and DigiBnd

ThickGateOx marks the 3.3 V gate oxide; DigiBnd selects the *digital* well rules (`NW.d1.dig` at 0.31 µm instead of `NW.d1` at 0.62). TGO is drawn on the cell boundary grown by 0.27 µm

(`TGO.a`) in x and **0.42 µm in y**: the shared rail's Activ crosses the cell boundary by 0.15 µm, and a plain 0.27 µm margin leaves only 0.12 µm of TGO past it at an *unabutted* row edge — a violation that appears exactly twice on any N-row test array (proven by stacking two and three rows: the count stays 2 and moves to the new outer edge). With 0.42 µm the cells are correct even at a block edge, while interior rows still merge far under the 0.86 µm `TGO.e` distance.

6.5 The five Metal1 re-routes

`M1.e` (0.22 µm space when a line is ≥ 0.30 µm wide and the parallel run exceeds 1.0 µm) is a rule the retarget *manufactures*: the y-map thickens horizontal lines past 0.30 and the x-map lengthens runs past 1.0, so 0.18 µm gaps that were legal in the thin-oxide cells become illegal with nothing moved closer. Generic edge-trimming cannot fix them — every gap abuts a contact column or a pin pad — so each of the five sites has an explicit hand re-route, kept as data (`M1E_EDITS`) in the generator:

Cell	Edit
`dlygate4sd2_1`	A2 pin pad bottom raised 35 nm (gap 0.185 → 0.220)
`mux2_1`	pin pad bottom raised 20 nm (0.20 → 0.22)
`sdfbbp_1`	strap bottom raised, starting 0.16 µm past a 0.16 µm arm junction
`sdfbbp_1`	15 nm × 930 nm bite between two contacts, splitting a 2.33 µm parallel run into 0.79 + 0.45 µm
`and4_2`	X pin pad narrowed 0.80 → 0.76 µm

Every edit is verified in code before it ships: merged polygon count unchanged (no split nets), `M1.a` width, `M1.b` space/notch, `M1.c1` contact enclosure, and the 5 nm grid. The guards rejected two earlier versions of these edits — a flush cut that left a 0.106 µm diagonal width, and a bite on a line only 0.17 µm wide — and an unguarded first attempt at automatic trimming had turned 6 violations into 37 (off-grid edges, sub-minimum widths, broken enclosures). Pin shapes (`8/2`) move with the drawing where a pad is edited.

6.6 Netlist ↔ layout synchronisation

The drawn width is authoritative: forcing the layout to the computed value means nudging an Activ edge under a channel, making it non-rectangular, and the extractor then reports a fractional gate length ($L = 0.4508\,\mu$m). `work/sync_netlist_widths.py` instead brings SPICE and CDL to the drawn geometry — matching per *finger*, not per device, because a folded `ng=4` device draws four channels (`buf_4`'s four devices are six drawn gates) — and recomputes `as/ad/ps/pd` with the deck formulas. The antenna diodes follow the same principle: the y-map stretches the antenna cell, so the diode area and perimeter in the netlist follow the drawing. Widths are handled in metres end-to-end; a unit slip once wrote `w=2400000.000u`.

6.7 Site, abutment and track benchmark

The PDK's tech LEF routes horizontally on a 0.42 µm pitch and vertically on 0.48 µm; the thin-oxide library's `CoreSite` is 0.48 × 3.78 µm, making it an exactly **9-track** library. This library is placed on the same grids: a **17-track** site of 0.48 × 7.140 µm.

	`sg13g2_stdcell` (LV)	**`sg13g2_stdcell_hv`**	**ratio**
site	`CoreSite` 0.48 × 3.78	`CoreSiteHV` 0.48 × 7.140	—
height in 0.42 µm h-tracks	**9**	**17**	1.89×
cell width quantum	0.48 µm	0.48 µm	—
mean cell width	—	1.52× LV	—
mean cell area (66 pairs)	—	—	**2.87×** (median 2.83, range 1.89–4.21)

Abutment holds. Every cell spans y = 0…7.140 with identical rail cross-sections, rails and rail Activ run through the cell edges, and the entire physical sign-off is performed on cells placed boundary-to-boundary with a mirrored second row sharing a rail — the configuration is not merely allowed but required, since rails, boundary-crossing Activ and ThickGateOx are only legal because neighbours continue them.

Neither grid property held automatically; both were imposed on the retargeted geometry:

- **17 tracks.** Device geometry alone (the 2.40× PMOS band plus the thick-oxide well clearances) gave 6.910 µm = 16.45 tracks, and horizontal M1/M3 tracks would have misaligned across row boundaries. `TRACK_PAD` adds the missing 0.230 µm in the mid-cell dead zone — the same place the `NW.d1` clearance opened, so every rule it touches only gains margin. The stretch carried three vertical Metal1 runs past `M1.e`'s 1.0 µm parallel-run threshold; each got a hand re-route in `M1E_EDITS` (a pin pad gives up 20–40 nm), verified by the same in-code guards.
- **Site-quantized widths.** The piecewise-linear x-map stretches only around gates and leaves gate-free spans unscaled, so mapped widths shared no quantum beyond the 5 nm grid. `pad_to_site` pads each cell's right boundary up to the next 0.48 µm multiple and extends with it exactly the shapes that continue across an abutment — rails, rail Activ, tap implants, wells, markers — so no device and no signal wire changes. Total padding across the library: 13.75 µm, ≈ 0.21 µm per cell.

The 3.3 % height cost of the 17th track and the ~2 % width cost of site padding are both inside the 2.87× mean area factor above.

7 Physical sign-off

7.1 Methodology

Cells are checked in an abutted context, never standalone: rails run off both cell ends, rail Activ crosses the boundary, and ThickGateOx merges across cells — all legal *because* cells abut. `work/make_drc_top.py` builds a row of all 66 cells with a second row mirrored above so the two share a rail; `work/make_drc_rows.py` generalises to N rows for the edge-artifact experiment.

Three measurement rules, each bought with a wrong conclusion:

- **One fixed invocation.** The PDK runner executes different rule decks depending on its flags (`--no_density` runs a monolithic deck emitting 4 files; the default emits 5; another flag set emits ~40). Counts from different invocations are not comparable — an interim draft of the documentation claimed "13 violations" by comparing across flag sets.
- **Diagnose from flat replication, not marker text.** In deep mode the `lyrdb` marker coordinates are in per-variant *cell-local* frames; read as top-cell µm they attributed all remaining violations to four cells that had nothing wrong with them. Every diagnosis that converged came from re-implementing the failing rule flat with the `klayout.db` API (`enclosed_check`, `separation_check`, `width_check`), which is dbu-exact.
- **Run the control.** IHP's own `sg13g2_stdcell` was pushed through the identical harness and invocation. That separated inherited flow artifacts (the six density items; the fill-cell LVS behaviour) from real defects in a single run, in both directions.

The verdict never comes from an exit code: both PDK runners exit 0 on failure, so pass/fail is parsed from the log and the report database.

7.2 DRC result

Rule class	This library	IHP control
cell rules	**0**	0
metal density (`M2.j M3.j M4.j M5.j TM1.c TM2.c`)	6	6

The path there: 781 raw violations after the first full retarget, driven to zero real items through the mechanisms of section 6 — the largest single steps being the `Gat.b1` gate-spacing inserts (26 items), the `pSD.i1` rail insert (104 items, located by measuring the actual error edges after a first misdiagnosis), fixed-size cut handling (`V1.a`), the channel-frozen slab alignment (`Act.c`/`Act.a`), the five `M1E_EDITS` re-routes (`M1.e`) and the 0.42 µm TGO y-margin (`TGO.a`).

7.3 LVS result

All 66 drawn cells match. The 62 cells with devices pass under the strict flags. The four `fill_*` cells contain no devices; klayout extracts a port-less empty circuit, the CDL subckt declares `VDD`/`VSS`, and the comparer reports the port-list difference — IHP's own fill cells fail the identical flow the same way. For exactly those four cells `work/run_lvs.sh` passes the runner's own `--ignore_top_ports_mismatch` relaxation; the port check is vacuous for a cell that extracts to nothing, while every cell with devices keeps the strict comparison.

LVS earlier caught five defect classes that dimensional checks could not — merged channels, collapsed notch devices, mis-paired folded fingers, a dropped substrate tie, stale diode geometry — which is why the per-cell sweep runs after every layout change.

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8 Liberty characterisation

`lib/sg13g2_stdcell_hv_typ_3p30V_25C.lib` is an NLDM library produced with **CharLib**, the open-source cell characteriser, at the 3.3 V / 25 °C typical corner. Every number in it comes from an **ngspice transient simulation of the shipped SPICE netlist** — the same `spice/sg13g2_stdcell_hv.spice` that LVS and the functional suites use, on the PSP103 Verilog-A device models loaded through OSDI. Nothing is estimated, and nothing is scaled from the thin-oxide library except the table *axes*. The final combinational run is 25 839 simulations, joined by a 2 962-task sequential run — 600 delay/slew tables over 66 cells, none empty — re-characterised after the layout work so the tables match the layout-synchronised netlists. When the layout later moved to the 17-track site the netlists stayed byte-identical — the pad only stretches dead zone and boundary — so the timing tables remained valid as-is; the one characterisation input that did change, cell **area**, was re-derived from the drawn boundaries (site-padded width × 7.140 µm) for all 52 cells, replacing the pre-layout estimates the first run had used.

8.1 Configuration

`work/gen_charlib_config.py` builds the CharLib YAML from the thin-oxide Liberty, which is trusted for everything the transform did not touch: the cell list, each cell's boolean function, pin directions and state declarations. Functions are translated by a recursive-descent parser (`work/boolexpr.py`) — XOR expanded to its AND/OR form — and **every** translated expression is checked against the original by truth-table equivalence before it is written. Logic thresholds are the standard 20/80 % slew and 50 % delay points. The 7 × 7 slew/load index grids are the thin-oxide grids rescaled by the measured 2.66× delay and 2.20× capacitance ratios (section 2), so the tables cover the same electrical territory as the original library rather than an arbitrary range.

8.2 What ngspice actually simulates

For each cell, each input→output arc, each rise/fall direction and each of the 7 × 7 slew/load points, CharLib drives the cell's subcircuit with a ramped input at that slew into that capacitive load and runs a transient analysis; propagation delay and output transition come from `.meas` statements evaluated on the waveforms. Per-cell leakage is a separate operating-point measurement per input state, and pin capacitance is a dedicated measurement biased **at**

both rails — a single mid-rail bias Miller-inflates the result (19.7 fF was measured for what is a 5.9 fF input) and was one of the first defects caught. Simulations run on all cores minus two with ngspice pinned to one thread each; oversubscription (ngspice's default four threads times a full worker pool) had slowed the first runs several-fold.

8.3 Making ngspice usable from CharLib

Five tool defects sit between CharLib and a correct library; each has a committed workaround in `work/`:

Defect	Symptom	Workaround
ngspice `-s` (server mode) ignores `.spiceinit`	OSDI never loads: "Unknown model type psp103va"	PATH shim (`ngspice-osdi-shim/`) injects `pre_osdi` + `pre_set num_threads=1` after the title line
`exec` inside the shim's pipeline replaces only the subshell	a second ngspice spinning at 98 % CPU	shim ends with an explicit `exit $?`
`ngspice-subprocess` backend never parses `.meas`	every timing table silently empty — while leakage, area and capacitance still populate	`ngspice-shared` backend (libngspice in-process) is mandatory
CharLib's leakage lookup lower-cases supply names; PySpice restores the netlist's spelling	`KeyError: 'vdd'` for any supply name	`charlib_patched.py` makes the branch-current lookup case-insensitive at runtime
emitted Liberty syntax defects	`function : "Y = !(A)"` left-hand sides, wrong `pulling_resistance_unit`, wrong `bus_naming_style`	`fix_lib.py` post-pass, run by `run_charlib.sh`

The shim was validated by comparing its measurements byte-for-byte against batch-mode ngspice on the same deck. The `.meas` defect deserves emphasis: it produces a structurally valid, plausible-looking library with empty timing tables, and nothing in the CharLib run reports an error.

8.4 Verifying the Liberty data

The library was loaded and exercised in OpenSTA early in the flow, but a loadable library is not a correct one, so the shipped `.lib` is verified **as data** by `work/verify_lib.py`, independently of how it was produced:

1. **Structure** — 66 cells, all 600 delay/slew tables populated, leakage on every cell. (Guards against the empty-`.meas` failure mode, which produces a structurally valid library with no timing in it.)
2. **Cross-view** — every lib cell and pin exists in the CDL under the same names.
3. **Physical** — every lib `area` equals the drawn boundary area of the GDS cell, to 1 nm².

4. **Monotonicity** — along every table's *load* axis, delay must not decrease. The load axis is taken from the template's `variable_1`/`variable_2` declaration, never assumed: this library declares `variable_1 : total_output_net_capacitance`, the **opposite axis order from IHP's thin-oxide library** (both are legal Liberty). The first version of this check assumed IHP's order, walked the slew axis, and reported 110 "violations" — including physically legitimate negative propagation delays at slow input slews, where the output crosses 50 % before the input does. A hasty hand point-check made the same axis mistake and manufactured a 22× discrepancy out of a (slew, load) pair that is not in the tables at all. Axis-aware, the real count is one outlier point: `xnor2_1`'s B-rise arc at (0.396 pF, 3.36 ns) reads 3.410 ns where a direct measurement gives 1.885 ns, smooth with its neighbours — the XNOR output glitches during the slow input traversal and CharLib's max-over-conditions latches the late crossing. The value ships as-is under a documented waiver in `verify_lib.py`: the error is pessimistic, the safe direction for timing sign-off, and hand-editing characterised data would be worse than a recorded exception.
5. **Measurement cross-check** — `inv_1`'s input capacitance against the independent both-rails ngspice measurement of `work/fo4.py` (5.87 fF).
6. **Sequential arcs** — every flip-flop and latch must carry an edge-triggered delay group and setup and hold constraint groups, and no constraint value may sit at the bisection search bounds (a pinned value means the pass/fail boundary was never bracketed — the fingerprint of both the skew-centering bug and the weak pass criterion) or outside –3…+5 ns. Result: **14/14 cells clean**.

Check 5 caught a real defect that had survived every earlier look: the pin capacitances were **6.8–7.5× too large** (`inv_1` 43.8 fF, `inv_16` 639 fF). CharLib's default `ac_sweep` procedure measures capacitance as a conductance slope at a floating DC bias, where Miller-multiplied Cgd dominates; the configuration now selects its `charge_integration` procedure — ramp the pin VSS→VDD→VSS and integrate the charge — which is exactly the both-rails method `fo4.py` validated, and reproduces the reference within ~10 % (6.0/6.7 fF fall/rise).

The delay tables themselves were confirmed correct by a point-check against an independent, hand-written ngspice transient at a table corner: (load 0.66 pF, slew 49.5 ps) reads **2.054 ns** in the table and measures **2.061 ns** in simulation — 0.3 % apart. The final gate on the shipped file: **PASS — 66 cells, 600 tables, 1 932 load-axis series monotone with the one documented waiver, Cin 6.44 fF vs the 5.87 fF reference (9.7 %), 14/14 sequential cells clean.**

8.5 Sequential cells

Stock CharLib 2.1.0 cannot characterize sequential cells at all, in three independent ways, each of which had to be discovered separately because every one fails silently or pathologically:

- Cells without `clock_slews` in their config raise a `KeyError` *inside* a generator that `omit_on_failure` swallows whole — zero simulations, zero errors, cells simply absent. This alone was the original "configured but produce nothing".

- `sequential_worst_case`, the clk→Q delay procedure, is an unimplemented stub: it yields tasks and returns the empty liberty skeleton (`# TODO`).
- The setup/hold contour procedure builds transients whose point count explodes on these cells (1.2 GB allocation requests); the failed allocation drives libngspice into its fatal “cannot recover” state and poisons the worker pool.

`work/seq_delay_procedure.py` replaces the last two through CharLib’s own procedure registry: `sequential_clk_to_q` measures clock/enable-to-output propagation and transition with a preload-pulse-then-measured-edge scheme, and `setup_hold_bisection` finds constraints by pass/fail bisection (~24 bounded simulations per constraint) with a **c2q-degradation criterion** — a trial passes only if the output arrives within 1.5× the nominal clk→Q, because final-state-only acceptance rides through luckily-resolved metastability and reports constraints pinned at the search floor. Latches are constrained against their **closing** edge; the first implementation left the enable asserted, the latch transparent, and every trial failing. Building this took twelve distinct defects found and fixed across the config contract, CharLib’s API, and the procedures themselves — the report’s companion history is in the git log; two are worth naming as measurement lessons: a skew-centering error granted every trial an extra half data-ramp of setup (pinning slow-slew rows at the bisection floor exactly like the criterion bug), and undriven set/reset pins let `rshunt` drift an active-low reset asserted, which held the flop cleared and turned a failed measurement into an infinite pool hang.

Validated behaviour (dfrbp_1, 3.3 V): setup +0.24…+0.59 ns rising with data slew and falling with clock slew; hold –0.43…–0.01 ns; clk→Q 0.49…2.6 ns monotone in load, with the metastability onset visible as c2q degradation near the setup boundary. Latch (dlhq_1): closing-edge setup between +0.3 and +1.0 ns, transparent D→Q 0.87 ns. Sequential leakage is a single-settled-state measurement (`work/seq_leakage.py`, 0.10–0.33 nW), unlike the all-states enumeration of the combinational cells — stated here because a reader comparing leakage groups would otherwise wonder.

8.6 Coverage

66 of 84 cells: 52 combinational, 9 flip-flops, 5 latches. Twelve cells are excluded by configuration with stated reasons — no output pin and no arcs (`fill_*`, `decap_*`, `antennanp`, `sighold`), no timing tables in the thin-oxide reference either (`tiehi`, `tielo`), statetable-based clock gates CharLib has no input form for (`lgcp_1`, `slgcp_1`). A further 20 — all 14 flip-flops/ latches and all 6 tri-state cells — are configured but produce nothing: CharLib emits an empty library for them without a single simulation and without reporting an error, so under `omit_on_failure` they vanish silently; the run log does not mention them. The `.lib` therefore cannot be used to synthesise sequential logic as it stands.

9 Verification summary

Every check run on the shipped library, in one place. “Golden reference” means the thin-oxide library or an independent hand measurement; nothing below is validated only against its own generator.

Check	Method	Scope	Result
Parasitic formulas	recompute vendor `as/ad/ps/pd`	all 924 thin-oxide devices	PASS, worst error $3.75{\times}10^{-4}$
Combinational logic	ngspice, both libraries in one deck vs thin-oxide golden	60 cells, 452 vectors, 12 high-Z states	**PASS** (re-run on final netlists)
Sequential logic	ngspice, clocked counter walk from common reset	16 stateful cells, 400 samples	**PASS** (re-run on final netlists)
Three-view consistency	xschem netlisting through symbols vs SPICE vs CDL	84 cells, 924 devices, field-by-field	**PASS**
DRC	PDK klayout deck, fixed invocation, abutted 2-row array	66 drawn cells	**0 cell-rule violations**; 7 chip-level density items
DRC control	identical harness on IHP `sg13g2_stdcell`	84 thin-oxide cells	0 cell rules, 6 density — density is the harness, not the cells
`M1.j` quantification	density over marker bbox vs cell area	padded 17-track array	35.0 % vs 35 % floor over bbox; 37.1 % over cell area; control 45.2 %
`TGO.a` row experiment	2-row vs 3-row array	outer-edge artifact hypothesis	count constant at 2, moves with the edge → artifact; then eliminated by the 0.42 µm y-margin
LVS	PDK klayout deck, per cell	66 drawn cells vs CDL	**66/66** (4 device-less fills need `--ignore_top_ports_mismatch`; IHP’s own fills fail identically without it)
LEF	klayout parse-back + pin sets vs CDL	66 macros, `CoreSiteHV`	parses; **66/66 pin sets match**; heights/widths asserted on the site grid
Liberty structure	`verify_lib.py` check 1	66 cells	600/600 tables populated, leakage on every cell
Liberty cross-view	check 2	all lib pins vs CDL	PASS
Liberty areas	check 3	54 cells with GDS	exact to 1 nm²

Check	Method	Scope	Result
Liberty monotonicity	check 4, load axis from the template declaration	1 932 delay series	1 931 monotone + 1 documented waiver (`xnor2_1`, table 3.410 ns vs 1.885 ns measured, pessimistic)
Liberty Cin	check 5 vs independent both-rails measurement	`inv_1` pin A	6.44 fF vs 5.87 fF — 9.7 % (this check caught the 6.8–7.5× `ac_sweep` Miller defect)
Liberty delay point-check	hand-written ngspice transient at a table corner	`inv_1` (0.66 pF, 49.5 ps)	table 2.054 ns vs simulated 2.061 ns — 0.3 %
Sequential arcs	check 6	14 flip-flops and latches	**14/14 clean**: delay + setup + hold groups present, no value pinned at search bounds
Sequential physics probe	direct-harness bisection boundary	`dfrbp_1`, `dlhq_1`	setup boundary bracketed with visible c2q degradation; latch closing-edge boundary bracketed; trends correct in slew and load
STA load test	OpenSTA	full library	loads and evaluates (early-flow validation of the identical pipeline)
Site geometry	boundary scan of the GDS	all 66 cells	uniform 7.140 µm = 17 tracks; every width a 0.48 µm site multiple

What is *not* verified, stated as plainly: no block has been placed and routed with the LEF; the 6 tri-state cells have no Liberty timing; the sequential leakage is a single-state number; and nothing here is silicon.

10 Known limitations

- **18 cells have no layout**: the eight D-flip-flop variants (`dfrbp_*`, `dfrbpq_*`, `sdfrbp_*`, `sdfrbpq_*`) run NMOS Activ up to the library channel cut, and ten others (`dlhq_1`, `dlhr_1`, `dllr_1`, `dllrq_1`, `ebufn_2`, `ebufn_8`, `lgcp_1`, `sighold`, `tiehi`, `tielo`) run PMOS Activ into the VDD rail; either way the band cannot scale without breaking the shared row height or rail abutment. They have full netlist, schematic, symbol and Verilog views, but no GDS.
- **The P&R view is untested by a P&R run.** LEF abstracts exist and are verified against the GDS and CDL, but no block has been placed and routed with them; OpenROAD/yosys flow integration is future work.

- **The 6 tri-state cells are absent from the Liberty.** CharLib's function model cannot express a high-impedance output and there is no registry work-around. They are simulation-verified (all 12 high-Z states) but carry no timing tables.
- **Sequential leakage is a single-settled-state measurement**, unlike the all-states enumeration of the combinational cells.
- **Delay cells shifted**: `dlygate4sd2_1` and `o21ai_1` had gate lengths below the thick-oxide minimum; their delay ratios differ from the thin-oxide originals. The decap cells store less per unit area — thicker oxide, same geometry.
- **Not silicon-proven, and not independently reviewed.** DRC/LVS-clean means the checks pass, not that the cells are known good in fabrication.

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11 Reproducing the library

```
cd /foss/designs/sg13g2_stdcell_hv/work

python3 gen_hv_lib.py            # netlists, schematics, symbols, Verilog
python3 gen_gallery.py           # sch/xschem/sg13g2_hv_stdcells.sch
python3 layout_retarget.py       # gds/ (66 cells; prints the 18 skips)
python3 sync_netlist_widths.py   # SPICE + CDL follow the drawn geometry
python3 gen_lef.py               # lef/ (CoreSiteHV + 66 macros)
python3 make_drc_top.py          # abutted two-row DRC context

python3 verify_logic.py          # combinational equivalence vs thin-oxide
python3 verify_seq.py            # sequential equivalence
python3 verify_sch.py            # three-view consistency
./run_lvs.sh                     # per-cell LVS, 66/66
# DRC: PDK run_drc.py on drc/drc_top.gds, default flags

./run_charlib.sh ../lib/sg13g2_stdcell_hv_typ_3p30V_25C.lib
# sequential cells: custom procedures registered by charlib_patched.py
#   (run with -f over the ff/latch names), then merge + leakage:
python3 merge_lib.py ../lib/sg13g2_stdcell_hv_typ_3p30V_25C.lib <seq.lib>
python3 seq_leakage.py
python3 verify_lib.py           # gates the shipped Liberty as data
```

Every number in this report is reproducible from these scripts; the `work/` directory is the provenance of record.